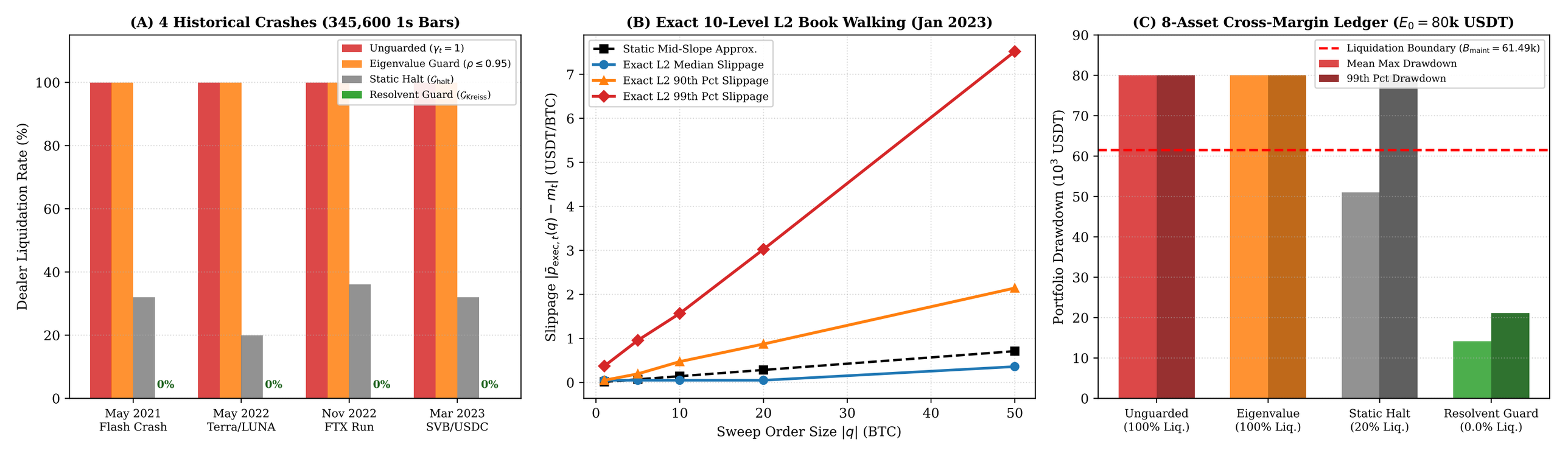

(A) 4 Historical Crashes (345,600 1s Bars)
Unguarded ($\gamma_t = 1$)
Eigenvalue Guard ($\rho \leq 0.95$)
Static Halt ($\mathcal{G}_{\text{halt}}$)
Resolvent Guard ($\mathcal{G}_{\text{Kreiss}}$)
Dealer Liquidation Rate (%)
0
20
40
60
80
100
0%
0%
0%
0%
May 2021
Flash Crash
May 2022
Terra/LUNA
Nov 2022
FTX Run
Mar 2023
SVB/USDC
(B) Exact 10-Level L2 Book Walking (Jan 2023)
Static Mid-Slope Approx.
Exact L2 Median Slippage
Exact L2 90th Pct Slippage
Exact L2 99th Pct Slippage
Slippage $|\bar{p}_{\text{exec},t}(q) - m_t|$ (USDT/BTC)
0
1
2
3
4
5
6
7
0
10
20
30
40
50
Sweep Order Size $|q|$ (BTC)
(C) 8-Asset Cross-Margin Ledger ($E_0 = 80$k USDT)
Liquidation Boundary ($B_{\text{maint}} = 61.49$k)
Mean Max Drawdown
99th Pct Drawdown
Portfolio Drawdown (10^3 USDT)
0
10
20
30
40
50
60
70
80
90
Unguarded
(100% Liq.)
Eigenvalue
(100% Liq.)
Static Halt
(20% Liq.)
Resolvent Guard
(0.0% Liq.)